INCH-POUND

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PERFORMANCE SPECIFICATION

CORROSION PREVENTIVE COMPOUND, SOLVENT-DISPERSED, AMBIENT TEMPERATURE DRIED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers solvent-dispersed corrosion preventive compounds that are deposited as thin films and dried at ambient temperatures by solvent evaporation. These products are formulated to be removed by solvent, oil, or steam; or to provide persistent, long-term, corrosion-control performance on metallic substrates.

1.2 Classification. Corrosion preventive compounds covered by this specification are furnished in the following classes, grades, and applications, as specified (see 6.2).

1.2.1 Classes. The classes of corrosion preventive compounds are as follows:

a. Class I – A corrosion preventive compound having a Volatile Organic Compound (VOC) content greater than 2.8 pounds per gallon (340 grams per liter) and less than or equal to 4.6 pounds per gallon (550 grams per liter).

b. Class II – A corrosion preventive compound having a VOC content greater than 0.8 pound per gallon (100 grams per liter) and less than or equal to 2.8 pounds per gallon (340 grams per liter).

c. Class III – A corrosion preventive compound having a VOC content less than or equal to 0.8 pound per gallon (100 grams per liter).

1.2.2 Grades. The grades of corrosion preventive compounds are as follows (see 6.1):

a. Grade 1 – Temporary, hard film, solvent-removable compound for corrosion protection during outdoor storage. NATO Code Number C-632 cites this specification to define performance requirements for this grade of corrosion preventive compounds.

b. Grade 2 – Temporary, soft film, solvent-removable compound for corrosion protection on equipment where miscibility with lubricants is required. NATO Code Number C-620 cites this specification to define performance requirements for this grade of corrosion preventive compounds.

c. Grade 3 – Temporary, water displacing, soft film, solvent-removable compound for corrosion protection during outdoor or indoor storage.

d. Grade 4 – Temporary, transparent, non-tacky film, solvent-removable compound for corrosion protection during outdoor or indoor storage.

e. Grade 5 – Temporary, low-pressure steam- or hot water-removable compound for corrosion protection during outdoor or indoor storage.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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f. Grade 6 – Temporary, solvent-removable compound for corrosion protection during storage of equipment (e.g., gear boxes) where miscibility with lubricants and nominal heat resistance are required.

g. Grade 7 – Temporary, solvent-removable compound for corrosion protection during indoor storage of machinery, instruments, or parts.

h. Grade 8 – Persistent, preservative compound for corrosion protection of metals in outdoor or indoor environments.

1.2.3 Applications. The corrosion preventive compound applications are as follows:

a. Application A – For brush application.

b. Application B – For dip application.

c. Application C – For fill and drain application.

d. Application D – For spray application using spray equipment.

e. Application E – For spray application using a self-pressurized container.

f. Application F – For applications with inert filler(s) or binder(s).

1.3 Part or Identifying Number (PIN). PINs to be used for corrosion preventive compound acquired to this specification are created as follows:

M	**16173**	-	**X**	-	**X**	-	**X**	-	**X**	-	**X**
Prefix for Military Specification	**Specification Number**	-	**Class (see code below)**	-	**Grade (see code below)**	-	**Application (see code below)**	-	**Container Size (see code below)**	-	**Color**

Class	Code
I	1
II	2
III	3

Grade	Code
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8

Application	Code
A	A
B	B
C	C
D	D
E	E
F	F

Container	Code
1-pint	1
1-quart	2
1-gallon	3
5-gallon	4
55-gallon	5
½- to 1-pint self-pressurized container	6
Cartridge (each cartridge to be individually wrapped with static mixer attached to cartridge)	7
Roll (roll to be individually wrapped)	8
Wipe (wipe to be individually packaged and determined quantity contained in single box)	9

Color
Manufacturer color indicated on commercial product data sheet

Example: M16173-1-1-A-1-[color]

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2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

O-M-232	-	Methanol (Methyl Alcohol)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680	-	Degreasing Solvent
MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-9000	-	Lubrication Oil, Shipboard Internal Combustion Engine, High-Output Diesel
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-PRF-22262	-	Abrasive Blasting Media, Ship Blast Cleaning

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 60, Appendix A-7, Method 24	-	Determination of Volatile Matter Content, Water Content, Density, Volume Solids, and Weight Solids of Surface Coatings
40 CFR 61	-	National Emissions Standards for Hazardous Air Pollutants
40 CFR 82	-	Protection of Stratospheric Ozone

(Copies of these publications are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This publication is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM A1008/A1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM B122/B122M - Standard Specification for Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar

ASTM B209/B209M - Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate

ASTM D93 - Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester

ASTM D609 - Standard Practice for Preparation of Cold-Rolled Steel Panels for Testing Paint, Varnish, Conversion Coatings, and Related Coating Products

ASTM D1640/D1640M - Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings

ASTM D1748 - Standard Test Method for Rust Protection by Metal Preservatives in the Humidity Cabinet

ASTM D3278 - Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus

ASTM D3359 - Standard Test Methods for Rating Adhesion by Tape Test

ASTM D4057 - Standard Practice for Manual Sampling of Petroleum and Petroleum Products

ASTM D5402 - Standard Practice for Assessing the Solvent Resistance of Organic Coatings Using Solvent Rubs

ASTM F483 - Standard Practice for Total Immersion Corrosion Test for Aircraft Maintenance Chemicals

ASTM G155 - Standard Practice for Operating Xenon Arc Lamp Apparatus for Exposure of Materials

(Copies of these publications are available online at www.astm.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS4377 - Magnesium Alloy, Sheet and Plate, 3.0Al - 1.0Zn – 0.20Mn (AZ31B-H24), Cold Rolled, Partially Annealed

(Copies of this publication are available online at www.sae.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC-SP 10/NACE No. 2 - Near-White Metal Blast Cleaning

(Copies of this publication are available online at www.ampp.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The corrosion preventive compound furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). The qualified class, grade, and application of corrosion preventive compounds shall be based on demonstrated corrosion-control performance without causing damage to the specific substrates to which they are applied (see 6.1).

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3.2 Material. The corrosion preventive compound shall be composed of a nonvolatile base material dispersed in a vehicle including but not limited to solvents, petroleum compounds, water or binders, fillers, or other residual materials conforming to the appropriate requirements in this specification. The corrosion preventative compound, with the exception of application F, shall be a homogeneous fluid visually free from sediment and suspended matter. Application F compounds shall consist of a grade 8 qualified, liquid corrosion preventative compound that is blended with inert fillers or binders.

3.2.1 Toxicity. When evaluated in accordance with 4.5.2.1, the corrosion preventive compound and any residual vehicle, binder, or filler shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5.2.1 and 6.4).

3.2.2 Prohibited materials. The material shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2. The material shall contain no class I or II Ozone Depleting Substances as specified in 40 CFR 82. The material shall contain no Hazardous Air Pollutants (HAPS) as defined in 40 CFR 61.

3.3 Volatile organic compound (VOC) content. When tested in accordance with 4.5.3, the VOC content for class I compounds shall be greater than 2.8 pounds per gallon (340 grams per liter) and less than or equal to 4.6 pounds per gallon (550 grams per liter), the VOC content for class II compounds shall be greater than 0.8 pounds per gallon (100 grams per liter) and less than or equal to 2.8 pounds per gallon (340 grams per liter), and the VOC content for class III compounds shall be less than or equal to 0.8 pounds per gallon (100 grams per liter).

3.4 Discernibility. When tested in accordance with 4.5.4, the compound film shall be visually discernible under standard laboratory lighting as a continuous, translucent, or opaque film with the color in accordance with the manufacturer's commercial product data sheet.

3.5 Film thickness. When tested in accordance with 4.5.5, the maximum film thickness shall be 4±2 mils (100±50 micrometers [µm]) for grade 1; 2±1 mils (50±25 µm) for grades 2, 4, and 6; 1±0.5 mils (25±13 µm) for grades 3, 5, and 7; and 6±2 mils (150±50 µm) for grade 8.

3.6 Storage stability. When tested in accordance with 4.5.6, the compound shall meet the requirements of 3.4, 3.5, 3.7, 3.10, and 3.12 for the applicable class, grade, and application.

3.7 Low-temperature storage stability. When tested in accordance with 4.5.7, the compound shall show no gelling, solidification, haze, or signs of a precipitate after visual inspection of the tilted test tube.

3.8 Corrosivity. When tested in accordance with 4.5.8, the metal samples shall show no evidence of pitting, etching, or darkening after exposure to the compound. All weight changes shall be less than or equal to the amount shown below.

Metal	Milligrams per square inch (milligrams per square centimeter)
Magnesium	3.2 (0.5)
Aluminum	1.3 (0.2)
Steel	1.3 (0.2)
Copper Nickel	1.3 (0.2)

3.9 Flash point. When tested in accordance with 4.5.9, the flash point shall be greater than or equal to 100 °F (38 °C) for all grades except grade 6, and greater than or equal to 140 °F (60 °C) for grade 6.

3.10 Miscibility with lubricating oil (grades 2 and 6). When tested in accordance with 4.5.10, the compound shall show no separation or sediment in any test tube.

3.11 Removability after exposure (excluding grade 8).

3.11.1 Grade 1 only. When tested in accordance with 4.5.11.1, the compound shall be completely removed over an area at least ¼ inch (0.64 centimeter) wide and the entire length of the stroke.

3.11.2 Grades 2, 3, 4, 6, and 7. When tested in accordance with 4.5.11.2, the compound shall be completely removed over an area at least ¼ inch (0.64 centimeter) wide and the entire length of the stroke.

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3.11.3 Grade 5 only. When panels are prepared in accordance with 4.5.11.3 and the testing apparatus is in accordance with 4.5.11.4, the compound shall meet the requirements of 3.11.3.1 and 3.11.3.2.

3.11.3.1 Hot water removability. When tested in accordance with 4.5.11.5, the level of residue remaining shall be not greater than 0.45 gram per square foot (4.84 grams per square meter).

3.11.3.2 Low-pressure steam removability. When tested in accordance with 4.5.11.6, the level of residue remaining shall be not greater than 0.15 gram per square foot (1.61 grams per square meter).

3.12 Salt spray exposure. When tested in accordance with 4.5.12, no panel shall show more than a trace of corrosion on the test surface of the panel. A trace of corrosion is defined as not greater than three corrosion spots, none of which is larger than 0.04 inch (1 millimeter) in diameter. Only corrosion within significant areas in accordance with ASTM D1748 shall be considered. Upon removal from testing, the panels shall be inspected in accordance with ASTM D1748 and shall be considered satisfactory if there is no etching, pitting, or other evidence of more than a trace of corrosion on the significant areas. Specimens that exhibit checking, alligatoring, or other irregularities of the corrosion preventive compound film are considered satisfactory if there is no etching, pitting, or other evidence of more than a trace of corrosion on the significant areas of the panel.

3.13 Accelerated weathering (grades 1 and 8). When tested in accordance with 4.5.13, no panel shall show more than a trace of corrosion on the test surface of the panel as defined in 3.12. Upon removal from testing, the panels shall be considered satisfactory if there is no etching, pitting, or other evidence of corrosion on the significant areas of the panels covered by the compound in accordance with ASTM D1748. Specimens that exhibit checking, alligatoring, or other irregularities of the corrosion preventive compound film are considered satisfactory if there is no etching, pitting, or other evidence of more than a trace of corrosion on the significant areas of the panel.

3.14 Flow resistance (grades 1, 3, and 7). When tested in accordance with 4.5.14, the compound flow towards the reference line shall not exceed 1/32 inch (0.8 millimeters).

3.15 Low-temperature adhesion. When tested in accordance with 4.5.15, there shall be no flaking of the corrosion preventive compound film within the areas bounded by the scratches. Chipping of the corrosion preventive compound extending more than 1/32 inch (0.8 millimeter) from either side of the scratch marks shall be considered flaking.

3.16 Drying (grades 1 and 4 only). When tested in accordance with 4.5.16, grade 1 compound shall be dry-to-touch within 4 hours, meaning the film does not adhere to the finger or rub up when the surface is touched.

3.16.1 Tackiness (grade 4 only) (applications A, B, C, D, and E only). When tested in accordance with 4.5.16.1, after 24 hours, the film shall be tack free exhibiting no residue on the finger nor any impression on the surface of the film when a fingertip is pressed lightly, without twisting, against the film.

3.17 Water displacement and water stability (grades 3, 5, and 7). When tested in accordance with 4.5.17, the test panels shall show no evidence of corrosion, mottling, or other abnormal surface stains or defects and shall displace water after storage in contact with water and when diluted in a 1-to-1 ratio with commercial petroleum “Base Oil 20 Blank”.

3.18 Low-temperature spraying (applications D and E only). When tested in accordance with 4.5.18, the compound shall spray uniformly without fingering or spitting and shall uniformly wet the steel test panels. When sprayed, the applied compound shall not exhibit froth or bubbling that results in compound running off of the steel test panels.

3.19 Flow after elevated temperature aging (grade 6 only). When tested in accordance with 4.5.19, the corrosion preventive compounds sample shall flow to wet the jar lid. There shall be visual evidence of the compound wetting the interior of the jar lid.

3.20 Transparency (grade 4 only) (applications A, B, C, D, and E only). When tested in accordance with 4.5.20, the compound shall be transparent and scribe-markings shall be legible following the exposure durations.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the Naval Sea Systems Command (NAVSEA) on sample units produced with equipment and procedures normally used in production. Qualification inspection shall consist of all of the examinations specified in table I, as applicable to the product grade.

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TABLE I. Qualification and conformance tests. [1/] [2/]

Characteristic	Product grade								Requirement paragraph	Test paragraph	Conformance test
	1	2	3	4	5	6	7	8			
Material	X	X	X	X	X	X	X	X	3.2	4.5.1	X
Toxicity	X	X	X	X	X	X	X	X	3.2.1	4.5.2.1	---
Prohibited materials	X	X	X	X	X	X	X	X	3.2.2	4.5.2.2	---
VOC content	X	X	X	X	X	X	X	X	3.3	4.5.3	X
Discernibility	X	X	X	X	X	X	X	X	3.4	4.5.4	---
Film thickness	X	X	X	X	X	X	X	X	3.5	4.5.5	---
Storage stability	X	X	X	X	X	X	X	X	3.6	4.5.6	---
Low-temperature storage stability	X	X	X	X	X	X	X	X	3.7	4.5.7	---
Corrosivity	X	X	X	X	X	X	X	X	3.8	4.5.8	---
Flash point	X	X	X	X	X	X	X	X	3.9	4.5.9	X
Miscibility with lubricating oil	---	X	---	---	---	X	---	---	3.10	4.5.10	X
Removability after exposure	X	X	X	X	X	X	X	---	3.11	4.5.11	---
Hot water removability	---	---	---	---	X	---	---	---	3.11.3.1	4.5.11.5	---
Low-pressure steam removability	---	---	---	---	X	---	---	---	3.11.3.2	4.5.11.6	---
Salt spray exposure	X	X	X	X	X	X	X	X	3.12	4.5.12	---
Accelerated weathering	X	---	---	---	---	---	---	X	3.13	4.5.13	---
Flow resistance	X	---	X	---	---	---	X	---	3.14	4.5.14	---
Low-temperature adhesion	X	X	---	X	---	X	---	X	3.15	4.5.15	---
Drying	X	---	---	---	---	---	---	---	3.16	4.5.16	---
Tackiness [3/]	---	---	---	X	---	---	---	---	3.16.1	4.5.16.1	---
Water displacement and water stability	---	---	X	---	X	---	X	---	3.17	4.5.17	---
Low-temperature spraying [4/]	X	X	X	X	X	X	X	X	3.18	4.5.18	---
Flow after elevated temperature aging	---	---	---	---	---	X	---	---	3.19	4.5.19	---
Transparency [3/]	---	---	---	X	---	---	---	---	3.20	4.5.20	---

NOTES:
1/ The test methods apply to all grades.
2/ Before tests are conducted, local cognizant industrial hygienists shall review operations and define health hazard control measures to limit worker exposure to potentially toxic materials, associated engineering controls, and required personal protective equipment.
3/ This test method applies to applications A, B, C, D, and E only.
4/ This test method applies to applications D and E only.

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4.3 Conformance inspection. When specified (see 6.2), conformance inspection shall consist of the conformance tests specified in table I and shall be performed on the first lot and every 10,000 gallons (38,000 liters) thereafter.

4.3.1 Lot. A lot shall consist of 10,000 gallons (38,000 liters) or 10,000 container units (e.g., cans, self-pressurized containers), whichever is less, of all the corrosion preventive compounds of the same class, grade, and application from a single uniform batch or a uniform blend of batches offered for delivery at one time.

4.3.2 Sampling for test. Sampling for conformance inspection shall be performed on the finished compound in accordance with ASTM D4057. From each lot, two separate 0.5- to 1-gallon (2- to 4-liter) samples or two separate container units shall be selected for conformance testing. One sample or container unit shall be used to conduct VOC content testing in accordance with table I. Another sample or container unit shall be used to conduct flash point testing in accordance with table I. For grade 1 corrosion preventive compounds, a third 0.5- to 1-gallon (2- to 4-liter) sample or separate container unit shall be selected for conformance testing. For grade 1, the third sample shall be used to conduct drying testing in accordance with table I. For grades 4 and 5 corrosion preventive compounds, a third 0.5- to 1-gallon (2- to 4-liter) sample or separate container unit shall be selected for conformance testing. For grades 4 and 5, the third sample shall be used to conduct miscibility with lubricating oil testing in accordance with table I.

4.4 Test panels. The panel material used for testing shall be cold-rolled mild steel in accordance with ASTM A1008/A1008M or as approved by NAVSEA. The test panels shall be approximately 2 by 4 by ⅛ inches (5.1 by 10.2 by 0.32 centimeters). The test panels shall be abraded with 220- to 240-grit aluminum oxide paper, cloth, or equivalent polishing medium. The test panels shall then be cleaned in accordance with procedure D of ASTM D609. For applications A, B, and C corrosion preventive compounds, the compound shall be applied by dipping the panel into a container of compound at normal laboratory temperatures of 72±5 °F (22±3 °C). For applications D and E corrosion preventive compounds, the compound shall be applied by spraying the compound in accordance with the manufacturer's commercial product data sheet at normal laboratory temperatures of 72±5 °F (22±3 °C). For application F corrosion preventive compounds, the compound shall be applied in accordance with the manufacturer's commercial product data sheet at normal laboratory temperatures of 72±5 °F (22±3 °C). The test panels shall be dried for a minimum of 24 hours in a draft-, dust-, and fume-free atmosphere at normal laboratory temperatures of 72±5 °F (22±3 °C) before further testing.

4.5 Test methods.

4.5.1 Material. Samples for material inspection shall be prepared by dispensing 1.7 to 2.0 U.S. fluid ounces (50 to 60 milliliters) of the corrosion preventive compound into an 8- by 1-inch test tube. The tube shall be placed in a rack at normal laboratory temperatures of 72±5 °F (22±3 °C) for 8 hours. At the end of this period, the sample in the tube shall be visually examined for conformance to 3.2.

4.5.2 Toxicity and prohibited materials.

4.5.2.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.1 as required by the qualifying activity. The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the material using data provided by the manufacturer/distributor to the NMCFHPC (see 3.2.1 and 6.4).

4.5.2.2 Prohibited materials. Prohibited materials shall be verified for conformance to 3.2.2 as required by the qualifying activity.

4.5.3 VOC content. The VOC content of the as-applied corrosion preventive compound shall be determined in accordance with 40 CFR 60, Appendix A-7, Method 24. VOC content shall be as specified in 3.3.

4.5.4 Discernibility. One panel shall be prepared and have the corrosion preventive compound applied in accordance with 4.4. The panel shall be stored at normal laboratory temperatures of 72±5 °F (22±3 °C) for 6 months for grades 1, 2, 4, 6, 7, and 8 and for 2 weeks for grades 3 and 5. After the allotted storage time, the corrosion preventive compound film on the panel shall be inspected in a well-lit and ventilated room. The results shall conform to 3.4.

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4.5.5 Film thickness. The average film thickness of the as-applied corrosion preventive compound shall be determined on a panel prepared as specified in 4.4. The surface shall be prepared and the compound applied in accordance with 4.4. The film thickness shall comply with 3.5 and shall be calculated using the following formula:

$$Film\ thickness\ (mils) = \frac{(1{,}000 \times \mathrm{W}) \times 0.061}{\mathrm{D} \times \mathrm{A}}$$

Where:

1,000 mil = 1 inch

W = weight of film (grams)

0.061 cubic inches = 1 cubic centimeter

D = density of the film (grams per cubic centimeter). Density is determined on the nonvolatile portion by means of a pycnometer.

A = total surface area of the panel coated with corrosion preventive compound (square inches).

4.5.6 Storage stability. Sufficient corrosion preventive compound to complete the required testing shall be stored for 1 year in a constant temperature chamber at 75±6 °F (24±3 °C). At the end of the 1-year period, the sample shall be visually inspected for the presence of cloudiness, sediment, suspended matter, or other changes in homogeneity. The stored sample shall be tested in accordance with table I for discernibility (see 4.5.4), film thickness (see 4.5.5), low-temperature storage stability (see 4.5.7), miscibility with lubricating oil (see 4.5.10), and salt spray (see 4.5.12) and the preservative shall meet the requirements of 3.6.

4.5.7 Low-temperature storage stability. Samples for storage stability shall be prepared by dispensing 1.7 to 2.0 U.S. fluid ounces (50 to 60 milliliters) of the corrosion preventive compound into an 8- by 1-inch test tube and placing a tightly fitting stopper in the tube. The tube shall be placed in a heated chamber at a temperature of 130±2 °F (54±1 °C) for 8 hours. At the end of this period, the tube and its contents shall be transferred immediately to a cold chamber maintained at -40±2 °F (-40±1 °C) and allowed to remain at that temperature for 16 hours. The test tubes shall not be agitated or subject to physical disturbance during the exposure in the cold chamber. This cycle shall be repeated four times, ending at the expiration of the last exposure at -40±2 °F (-40±1 °C). After removal from the cold chamber, the test tube shall be allowed to remain upright in a vertical orientation at a normal laboratory temperature of 72±5 °F (22±3 °C) for 24 to 36 hours. The test tube shall then be tilted to a horizontal position and held in that position for 5 seconds. The sample shall be visually examined for evidence of gelling, precipitate, haze, or separation. The results shall conform to the requirements of 3.7.

4.5.8 Corrosivity. Panels of the following metals shall be used in this test (see 6.6):

a. Magnesium, SAE AMS4377, AZ31B,

b. Aluminum 2024, ASTM B209/B209M,

c. Steel, ASTM A1008/A1008M, cold-rolled,

d. Copper nickel C70600, ASTM B122/B122M.

4.5.8.1 Panel preparation. Three panels of each of the four metal types designated in 4.5.8, with the geometry in accordance with ASTM F483, shall be polished to remove pits, burrs, and irregularities from all faces and edges with 220- to 240-grit aluminum oxide paper, cloth, or equivalent polishing medium. Fresh abrasive cloth or paper shall be used for each set of three panels of the four metal types. After polishing, panels shall be cleaned by swabbing with high-flash naphtha followed by a final rinse with a solution of 95 percent O-M-232 methanol and distilled water. The weight of each panel shall be recorded to the nearest milligram.

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4.5.8.2 Immersion. The four sets of three panels prepared in 4.5.8.1 shall be tested for corrosivity by total immersion of each test panel in the corrosion preventive compound in accordance with ASTM F483 with the following exceptions: the test temperature shall be 130±2 °F (54±1 °C) and the test duration shall be continuous immersion for a minimum of 7 days. Upon completion of the immersion period, the panels shall be inspected and cleaned. The panels shall be visually inspected for corrosion and any loose corrosion products shall be removed by swabbing with high-flash naphtha followed by a final rinse with a solution of 95 percent O-M-232 methanol and distilled water. After cleaning, the panels shall be visually inspected for evidence of corrosion pitting, darkening, or etching. After the panels are dry, each cleaned panel shall be weighed. The weight loss or gain per total panel surface area shall be calculated (i.e., the area of both sides of each panel) in milligrams per square inch or milligrams per square centimeter. The amount of weight change per unit area and results of the visual inspection shall comply with 3.8.

4.5.9 Flash point. The test for flash point for all grades shall be in accordance with ASTM D93, ASTM D3278, or as approved by NAVSEA. The results shall conform to the requirements of 3.9.

4.5.10 Miscibility with lubricating oil (grades 2 and 6). Three 3.381-U.S. fluid ounce (100-milliliter) cone-shaped centrifuge tubes shall be procured for testing each corrosion preventive compound grade. One tube shall be filled with 3.0 to 3.2 U.S. fluid ounces (89 to 95 milliliters) of lubricating oil in accordance with MIL-PRF-17331; one tube shall be filled with 3.0 to 3.2 U.S. fluid ounces (89 to 95 milliliters) of lubricating oil in accordance with MIL-PRF-9000; and the remaining tube shall be filled with 3.0 to 3.2 U.S. fluid ounces (89 to 95 milliliters) of lubricating oil in accordance with MIL-PRF-2104, 15W-40 grade. Each centrifuge tube shall have 0.14 to 0.17 U.S. fluid ounces (4 to 5 milliliters) of the corrosion preventive compound added. Each tube shall be stoppered and shaken by hand until the mixture appears uniform. The stopper shall be loosened, and each tube shall be placed in a beaker in an upright or vertical orientation. The beaker containing the centrifuge tubes shall be placed into an explosion-proof oven at a temperature of 170±2 °F (77±1 °C) for 15 minutes. Each tube shall be removed, and the content shall then be examined for separation or sediment. The contents shall be allowed to cool to a normal laboratory temperature of 72±5 °F (22±3 °C) and then examined again for separation or sediment. Each tube shall remain at normal laboratory temperatures for an additional 24 hours and then shall be examined again for separation or sediment. The compound shall conform to the requirements of 3.10.

4.5.11 Removability after exposure.

4.5.11.1 Grade 1. Two test panels shall be prepared in accordance with 4.4 and grade 1 compound shall be applied to each panel. One test panel shall be exposed for a minimum of 336 hours in accordance with 4.5.12. The other test panel shall be exposed for a minimum of 1,200 hours in accordance with 4.5.13. Following exposure, the compound shall be removed from the test panel using a cotton cloth saturated with dry-cleaning solvent conforming to type I of MIL-PRF-680 in accordance with ASTM D5402. The test panel shall not be cleaned with tap water prior to testing and no dry film thickness measurements are required. The length of the wiping stroke shall be 2 to 3 inches (5.1 to 7.6 centimeters) and each stroke shall represent a complete back and forth motion or a “double rub” over one area of the panel. The wiping process shall be accomplished at slight to medium pressure and solvent shall be added to the cotton cloth on an as-needed basis to maintain saturation of the cloth. At the completion of 100 double rub strokes, the panel shall be visually inspected for compound removal. The results shall conform to the requirements of 3.11.1.

4.5.11.2 Grades 2, 3, 4, 6, and 7. One test panel shall be prepared in accordance with 4.4 for each of the corrosion preventive compound grades (i.e., grades 2, 3, 4, 6, and 7) being tested. Each grade being tested shall be applied to one of the test panels. The test panel shall be stored for a minimum of 9 months at a normal laboratory temperature of 72±5 °F (22±3 °C) in a container that prevents the panel from being exposed to dust or solvent fumes. Following exposure, the compound shall be removed from the test panel using a cotton cloth saturated with dry-cleaning solvent conforming to type I of MIL-PRF-680 in accordance with ASTM D5402. The test panel shall not be cleaned with tap water prior to testing and no dry film thickness measurements are required. The length of the wiping stroke shall be 2 to 3 inches (5.1 to 7.6 centimeters) and each stroke shall represent a complete back and forth motion or a “double rub” over one area of the panel. The wiping process shall be accomplished at slight to medium pressure and solvent shall be added to the cotton cloth on an as-needed basis to maintain saturation of the cloth. At the completion of 100 double rub strokes, the panel shall be visually inspected for compound removal. The results shall conform to the requirements of 3.11.2.

4.5.11.3 Grade 5. Eight steel panels shall be fabricated from ASTM A1008/A1008M steel, or an equivalent mild steel grade approved by NAVSEA, that are ⅛ by 2 by 2 inches (0.32 by 5.1 by 5.1 centimeters) as shown on

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figure 1. As shown on figure 1, the panels shall have a ⅛-inch (0.32-centimeter) hole centered ⅛ inch (0.32 centimeter) from one edge. Panels shall be finished and cleaned as specified in 4.4, weighed to the nearest milligram, and the corrosion preventive compound applied as specified in 4.4. The test panels shall be aged in a draft-, dust- and fume-free indoor atmosphere for a minimum of 9 months before performing the hot water and low-pressure steam removability tests.

4.5.11.4 Removability apparatus. The apparatus for determining the hot water and low-pressure steam removability tests shall be as shown on figure 1 or as approved by NAVSEA. The apparatus shall consist of a rectangular box 17 by 2 inches (43.1 by 5.1 centimeters) wide by 2.25 inches (5.715 centimeters) deep with a removable cover and gasket. Welded to a hole in each end of the box shall be 1½-inch (3.8-centimeter) nominal pipe size (NPS) pipe fittings to which are attached a 1½-inch (3.8-centimeter) valve and 1½-inch (3.8-centimeter) flange. The cover shall contain lugs for holding nine test panels perpendicular to the cover and at a 45-degree angle to the direction of hot water or steam flow through the apparatus. A pressure gauge and a thermometer shall be fitted in the box to measure the temperature of the hot water or steam in the apparatus.

4.5.11.5 Hot water removability. The eight aged panels described in 4.5.11.3 shall be secured to the cover of the removability apparatus described in 4.5.11.4 (see figure 1) with an uncoated panel installed as a baffle plate at the bottom of the apparatus to regulate the flow and prevent direct impingement of hot water on the aged panels. The cover shall be attached to the box. The apparatus shall be mounted vertically with the valve at the bottom. The 180±5 °F (83±3 °C) hot water shall be recirculated through the box from the bottom at a rate of approximately 20 gallons per minute (76 liters per minute) for a minimum of 3 hours. Although the quantity of hot water used for this test is not critical, a 50-gallon (190-liter) reservoir is recommended to provide enough hot water at a consistent temperature for this test. After testing, the test apparatus shall be drained and disassembled. The baffle plate shall not be inspected or weighed. The test panels shall be removed, dried, and weighed to the nearest milligram. The amount of corrosion preventive compound remaining shall be calculated as follows:

$$Residue\ (g/ft^2) = 16(A - B)$$

Where:

A - Final weight of panel in grams.

B - Uncoated weight of panel in grams.

16/ft² - Conversion factor for the total area including the panel thickness.

The level of residue shall conform to the requirements of 3.11.3.1.

4.5.11.6 Low-pressure steam removability. The eight aged panels described in 4.5.11.3 shall be secured to the cover of the removability apparatus described in 4.5.11.4 (see figure 1) with an uncoated panel installed as a baffle plate at the top of the apparatus. The cover shall be attached to the box. The box shall be mounted vertically with the steam entering from the top. Wet steam at 25±1 pounds per square inch (1.76±0.07 kilograms per square centimeter) shall be piped to a 3/16-inch (0.476-centimeter) sharp-edged orifice placed in the flange at the top of the apparatus. The pressure in the apparatus shall be throttled to 15±1 pounds per square inch (1.05±0.07 kilograms per square centimeter) by the valve at the bottom and the steam allowed to flow through the apparatus for a minimum of 3 hours. After testing, the steam shall be shut off, the apparatus vented, and the test apparatus disassembled. The baffle plate shall not be inspected or weighed. The test panels shall be removed from the apparatus, dried, and weighed to the nearest milligram. The amount of corrosion preventive compound remaining shall be calculated as follows:

$$Residue\ (g/ft^2) = 16(A - B)$$

Where:

A - Final weight of panel in grams.

B - Uncoated weight of panel in grams.

16/ft² - Conversion factor for the total area including the panel thickness.

The level of residue shall conform to the requirements of 3.11.3.2.

4.5.12 Salt spray exposure. Three panels shall be prepared for each grade and corrosion preventive compound shall be applied in accordance with 4.4. No “stop-off” material or epoxy coating shall be applied to the test panel

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edges. The test panels shall be exposed for the minimum specified time in table II in accordance with ASTM B117. At the completion of the exposure period, the three panels shall be inspected for traces of corrosion. The panels shall conform to the requirements of 3.12.

TABLE II. Minimum salt spray exposure time.

Compound (grade)	Exposure time (hours)
1	336
2	168
3	168
4	336
5	168
6	168
7	72
8	504

4.5.13 Accelerated weathering (grades 1 and 8). Three panels shall be prepared and corrosion preventive compound shall be applied in accordance with 4.4. The panels shall be exposed to an accelerated weathering test. The apparatus shall be in accordance with ASTM G155 except that the water temperature shall be maintained at 70±10 °F (21±6 °C) and the pH of the water shall be 8±1. At the end of a minimum of 1,200 hours of exposure, the panels shall be evaluated for resistance to corrosion and conform to the requirements of 3.13.

4.5.14 Flow resistance (grades 1, 3, and 7). One steel panel shall be finished and cleaned in accordance with 4.4. A line shall be scribed using a sharp probe or knife parallel to, and ⅞ inch (2.22 centimeters) from, the bottom long edge of the panel as shown on figure 2, item (a). A 1-inch (2.5-centimeter) wide strip of masking tape shall be applied to both sides of the bottom half of the panel to cover the scribed line as shown on figure 2, item (b). The corrosion preventive compound shall be applied to the test panel as specified in 4.4. After the 24-hour drying period, the tape shall be carefully removed to provide a uniform edge between the compound and the exposed steel. The panels shall be suspended such that the edge of the 1-inch (2.5-centimeter) strip that has no compound is parallel to the long edge of the panel and below the compound edge. The line previously scribed on the panel shall now be ⅛ inch (0.32 centimeter) below, and parallel to, the bottom edge of the previously applied compound as shown on figure 2, item (c). The panel shall then be suspended for a minimum of 4 hours in a forced convection oven maintained at a temperature of 175±2 °F (79±1 °C). The panel shall be removed from the oven and allowed to cool to a normal laboratory temperature of 72±5 °F (22±3 °C). The distance between the compound edge that has flowed closest to the scribe line shall be measured with a machinist ruler that has 1/32-inch increments. The distance between the compound edge and the scribe line on the test panel shall conform to 3.14.

4.5.15 Low-temperature adhesion. One panel shall be prepared and have the corrosion preventive compound applied in accordance with 4.4. Following the allotted drying period in 4.4, the panel shall be cooled to a temperature of 0±2 °F (-17±1 °C) for grade 1 or -40±2 °F (-40±1 °C) for grades 2, 4, 6, and 8, and maintained at that temperature for a minimum of 1 hour. While the panels are at this temperature, the compound film shall be cut through with a sharp probe or knife in accordance with ASTM D3359, method B except that the cross hatch shall consist of four parallel scratches approximately ⅛ inch (0.32 centimeter) apart and 1 inch (2.5 centimeters) long. The scribed surface shall not be brushed in accordance with ASTM D3359, method B. Pressure-sensitive tape shall not be applied to the scribed surface in accordance with ASTM D3359, method B. The scribed surface shall be visually inspected in accordance with ASTM D3359, method B. The test results shall conform to the requirements of 3.15.

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4.5.16 Drying (grades 1 and 4 only). One test panel shall be prepared and have the corrosion preventive compound applied in accordance with 4.4 except that the panel shall be allowed to drain and dry in a vertical position at a normal laboratory temperature 72±5 °F (22±3 °C) for a minimum of 4 hours rather than the 24 hours designated in 4.4. After the minimum 4-hour drying time, the compound film shall be examined in accordance with ASTM D1640/D1640M for dry-to-touch time, drying oils, and meet the requirements of 3.16.

4.5.16.1 Tackiness (grade 4 only) (applications A, B, C, D, and E only). After completion of testing specified in 4.5.16, the panel shall dry for an additional 20 hours (24 hours total), after which the compound film shall conform to the tackiness requirements in 3.16.1.

4.5.17 Water displacement and water stability (grades 3, 5, and 7).

4.5.17.1 Preparation of test panels prior to abrasive blasting. Three test panels approximately 2 by 3 by 1/16 inches (5.1 by 7.6 by 0.16 centimeters) shall be prepared from ASTM A1008/A1008M carbon steel, or an equivalent mild steel grade approved by NAVSEA. Pitted or corroded steel shall not be used for the test panels. The surfaces and edges shall be uniformly abrasive blasted to an SSPC-SP 10/NACE No. 2 level of cleanliness using garnet media conforming to MIL-PRF-22262.

4.5.17.2 Preparation of test panels after abrasive blasting. After each panel is abrasive blasted, it shall be rinsed by being placed in a container of methyl alcohol conforming to grade A of O-M-232. The panel shall remain in the container for a minimum of 15 seconds. The test panel and methyl alcohol shall be warmed to a minimum of 5 degrees above the dew point of the room. To ensure removal of oil and grease contamination, the panels shall be removed from the methyl alcohol and when dried, the panels shall be rinsed in high-flash naphtha that has a flash point greater than or equal to 140 °F (60 °C) to a temperature that is a minimum of 5 degrees above the dew point of the room. Immediately after cleaning, the panels shall be stored at less than 50 percent relative humidity to preclude corrosion. The panels shall be handled with forceps or gloves at all times during and after the abrasive blasting.

4.5.17.3 Preparation of compound. Samples shall be prepared by placing 1.7±0.03 U.S. fluid ounces (50±1 cubic centimeters) of the compound in a 4.2-U.S. fluid ounce (125-cubic centimeter) Erlenmeyer flask and adding 0.17±0.003 U.S. fluid ounces (5±0.1 cubic centimeters) of distilled water to the flask. The flask shall be shaken to create an emulsion with a uniform appearance. The stoppered flask shall be stored overnight at 130±2 °F (54±1 °C) and cooled to 72±5 °F (22±3 °C) for testing.

4.5.17.4 Water displacement procedures. Each of three panels prepared as specified in 4.5.17.1 and 4.5.17.2 shall be dipped in distilled water momentarily and drained in a vertical position for not greater than 5 seconds, with the bottom edge in contact with absorbent paper. The panels shall then be immersed horizontally and without agitation for a minimum of 15 seconds in a glass or plastic dish containing the compound previously prepared as specified in 4.5.17.3. The panels shall be removed from the dish and excess compound shall be allowed to drip off the panels into the dish for a maximum of 15 seconds. The panels shall then be placed in a static humidity cabinet maintained at a relative humidity of 95±5 percent for a minimum of 1 hour at 72±5 °F (22±3 °C). After 1 hour in the static humidity cabinet, the panels shall be cleaned in accordance with 4.5.17.2 and examined for evidence of rust, mottling, or other abnormal surface stains and defects. The panels shall conform to the requirements of 3.17.

4.5.17.5 Dilution. At a 1-to-1 ratio, the compound shall be diluted with a commercial “Base Oil 20 Blank” at 100±5 °F (38±3 °C) and the water displacement test shall be performed as specified in 4.5.17.4. The test results shall conform to the requirements of 3.17.

4.5.18 Low-temperature spraying (applications D and E only). One steel panel shall be finished and cleaned according to the procedure of 4.4. The compound in the as-received, or ready to apply, container shall be conditioned for a minimum of 24 hours in a cold chamber maintained at 40±2 °F (4±1 °C). The container shall be removed from the cold chamber and an attempt shall be made to spray the corrosion preventive compound onto the panel. The spraying and compound film characteristics shall conform to the requirements of 3.18.

4.5.19 Flow after elevated temperature aging (grade 6 only). Samples shall be prepared by placing 2±0.2 U.S. fluid ounces (60±6 milliliters) of the corrosion preventive compound into a 4-fluid ounce (120-milliliter) capacity clear glass jar with an exterior diameter of 2.4±0.2 inches (6±0.5 centimeters) and a height of 2.6±0.2 inches (6.7±0.5 centimeters) with a screw top lid. The unsealed jar shall be exposed to a normal laboratory temperature of 72±5 °F (22±3 °C) for a period of 24±2 hours. The open jar shall be placed in an oven heated to 212±5 °F (100±3 °C) for a period of 72±2 hours. After removal from the oven, the unsealed jar shall be allowed to cool for a minimum of 24 hours at normal laboratory temperature of 72±5 °F (22±3 °C). After the jar has cooled,

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the lid shall be placed on the jar and the jar shall be oriented in a horizontal position such that the corrosion preventive compound will flow toward the jar lid for a period of 30±3 seconds. The flow of corrosion preventive compound in the jar shall be observed for conformance with the requirements of 3.19.

4.5.20 Transparency (grade 4 only) (applications A, B, C, D, and E only). One panel shall be prepared and have the corrosion preventive compound applied in accordance with 4.4 except that the panel shall be allowed to drain and dry in a vertical position at a normal laboratory temperature 72±5 °F (22±3 °C) for 4 hours. Prior to application of the corrosion preventive compound to the panel, one side of the panel shall be marked with five scribe markings, with no individual marking more than 3⁄16 inch tall or wide. These markings may be the numbers or letters normally used in identifying the panel. A well-sharpened scratch-awl should be adequate for marking. The panel shall be stored for 1 year at a normal laboratory temperature of 72±5 °F (22±3 °C) in a container that prevents the panel from being exposed to dust or solvent fumes. After the 4-hour drying period and after the 1-year exposure test duration, the scribe markings and corrosion preventive compound film on the panel shall be visually inspected in a well-lit and ventilated room. The results shall conform to the requirements of 3.20.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The corrosion preventive compounds covered by this specification are military unique systems manufactured to effectively control corrosion in military-unique applications like interiors of ship and submarine control surfaces, complex topside equipment housings, and mixed metal joints used to install combat systems. They are intended for use on spaces that are inaccessible for conventional coating installations and as temporary protective materials intended to prevent corrosion of military hardware during shipping and storage. Grades 2 and 6 corrosion preventive compounds define military-unique requirements for preservatives that can be removed from components like main reduction gears by simply flushing the units with the required lubricating oil. These corrosion preventive compounds have been formulated to comply with air pollution regulations that allow a range of VOC contents because these products are not to be thinned. The corrosion preventive compounds may be applied using a fill-and-drain process, by spray, or by using brushes/rollers to bare metallic substrates. They are intended for the uses specified in 6.1.1 through 6.1.8. With the exception of grade 8 corrosion preventive compound that is intended for persistent protection of metallic surfaces and application F for applications with inert filler(s) or binder(s), other corrosion preventive compounds procured to this specification are form, fit, and function replacements for prior revisions. Existing stocks of grades 1 through 5 corrosion preventive compounds procured to this specification are acceptable for use until depleted. All future requisitions should be to the current revision of this specification.

6.1.1 Grade 1. Grade 1 hard film, solvent-removeable corrosion preventive compound is used for temporary protection of metals when exposed for extended periods to outdoor weather conditions that is also cited as a NATO Code Number C-632 corrosion preventive compound.

6.1.2 Grade 2. Grade 2 soft film, solvent-removable, or removable by flushing with hot lubricating oil, corrosion preventive compound is used for temporary protection of interior or exterior metals or outdoor protection of material for limited periods where metal temperatures do not reach levels that produce prohibitive flow of the corrosion preventive film that is also cited as a NATO Code Number C-620 corrosion preventive compound.

6.1.3 Grade 3. Grade 3 soft film, solvent-removable corrosion preventive compound is used for temporary protection of metals where water or saline solution must be displaced from corrodible surfaces and the corrosion prevented or arrested.

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6.1.4 Grade 4. Grade 4 solvent-removable corrosion preventive compound is used for protection of machinery or instrument surfaces, bearings, or other materials with or without use of supplementary barrier materials, or for outdoor protection of machinery or parts.

6.1.5 Grade 5. Grade 5 hot water or low-pressure steam-removable corrosion preventive compound is used for temporary protection of metals when exposed to indoor or outdoor weather conditions and chemical methods cannot be used for removal.

6.1.6 Grade 6. Grade 6 solvent-removable and removable by flushing with hot lubricating oil corrosion preventive compound is used for temporary protection of metals where miscibility with lubricants and nominal heat resistance are desired.

6.1.7 Grade 7. Grade 7 soft film, solvent-removable corrosion preventive compound is used for temporary protection during indoor storage of machinery, instruments, or parts.

6.1.8 Grade 8. Grade 8 corrosion preventive compound is used for persistent protection of metals when exposed to indoor or outdoor weather conditions.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class, grade, and application required (see 1.2).

c. Container size (see 1.3).

d. When conformance inspection is required (see 4.3).

e. Packaging requirements and recommended packaging and labeling methods (see 5.1 and 6.7).

f. When a Safety Data Sheet (SDS) is required (see 6.5).

g. Batch VOC content certification (see 6.9).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 16173 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

6.5 Safety data sheets. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with appendix D of 29 CFR 1910.1200. In order to obtain the SDS, federal acquisition regulation clause 52.223-3 will be in the contract. The contracting activity should be given an SDS at the time of contract award.

6.6 Metal panels. Panels of the metals used in corrosion tests may be obtained from a commercial supplier including, but not limited to, one of the following: Metaspec Services LLC 355 Spencer Lane Ste 1-112, San Antonio, TX 78201 (CAGE: 992G4); Alabama Specialty Products, Inc., 152 Metal Samples Rd., P.O. Box 8, Munford, AL 36268 (CAGE: 0UW16); or ACT Test Panels LLC, 273 Industrial Dr., Hillsdale, MI 49242 (CAGE: 3Z8A5) (see 4.5.8).

6.7 Packaging recommendations. Suggested packaging and marking requirements are contained in tables III and IV (see 6.2).

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TABLE III. Packaging.

Packaging	Recommended requirements for direct Government acquisitions
Containers	(1) The corrosion preventive compounds should be specified to be furnished in cans capable of holding 1 pint (0.47 liter), 1 quart (0.945 liter), 1 gallon (3.78 liters), 5 gallons (18.9 liters), and 55 gallons (207.9 liters). In addition, compounds may also be furnished in cartridges, tapes, aerosol cans, pastes, putties, wraps, wipes, or other packaging formats as defined in the acquisition document. Application F compound containers should be specified to be furnished in container sizes measured in ounces (grams) or linear dimensions such as inches (centimeters) or feet (meters). (2) Multiple friction plug containers should be in accordance with PPP-C-96, type V, class 2. Compounds intended for interior service should be as specified therein. Compounds intended for exterior service in the weather should be specified. Wire handles as specified therein should be provided for the 1-gallon container. Closure of the properly filled and sealed cans should be as specified in the appendix thereto. (3) All containers should comply with the requirements of the Uniform Freight Classifications (UFC), the National Motor Freight Classification (NMFC), and the applicable requirements of the 49 CFR 178, Specifications for Packagings. (4) Unit of procurement: The products covered by this specification should be purchased by volume. The unit of procurement should be in multiples of 1 U.S. liquid gallon or 1 liter at 60 °F (15.5 °C), and the applicable requirements of 49 CFR 178, Specifications for Packagings.
Commercial packaging	(1) Commercial packaging should be in accordance with ASTM D3951. (2) All containers should comply with the requirements of the UFC, the NMFC, and the applicable requirements of 49 CFR 178, Specifications for Packagings.
Packing	Packing should be specified as follows: (1) Overseas delivery packing. Intermediate containers of like size containers of product should be in accordance with ASTM D6251/D6251M packed in close-fitting wood boxes. Box closure and strapping should be as specified in the applicable box specification or the appendix thereto except that strapping should be flat and the finish B. (2) Domestic delivery (level B) packing. Level B packing should be as for level A, except that boxes should be domestic type or class and the strapping should be finish A or B. (3) Commercial packing. The product in the unit can and intermediate containers should, as applicable, be packed in multiples of like sizes in accordance with UFC, NMFC, and 49 CFR 178 requirements.
Palletization	Containers should be palletized in accordance with MIL-STD-147. Only one size unit or container should be placed on a pallet.
Packing for acquisitions involving direct delivery to Navy ships or installations	Treated lumber and plywood. All lumber and plywood, including laminated veneer materials, used in shipping container and pallet construction, member, blocking, bracing, and reinforcing must be fire-retardant treated material in accordance with MIL-L-19140 as follows: (1) General use, weather resistant: MIL-L-19140, type II, category 1. (2) General use, non-weather resistant: MIL-L-19140, type I, category 1.
Safety Data Sheets (SDS) and product /procedure data sheets	A copy of the SDS and company product data/procedure sheets should be attached to the shipping document for each destination (see 6.5).
VOC content certification sheets	VOC content certification sheets for each batch of corrosion preventive compound must be provided when requested by the procuring activity (see 6.9).

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TABLE IV. Marking.

Marking type	Recommended marking
Bar codes	Marking should include bar codes.
Hazardous warnings	(a) Labels should be in accordance with 29 CFR Parts 1910, 1915, 1917, 1918, 1926, and 1928, as well as 49 CFR. (b) All individual containers should have the following marking: "CAUTION: This product contains volatile solvents and vehicles that can create hazardous vapor concentrations." (c) Each component container, shipping container, and palletized load should be marked with the appropriate hazardous symbol. (d) Unit containers should be marked: "This product is Asbestos, Lead, Chromium, Cadmium free, except for possible trace levels."
Volatile organic compound (VOC) content	"Contains (insert VOC content) grams per liter (insert VOC content in pounds per gallon) of volatile organic content per 40 CFR CH. 1, Part 60, Appendix A (EPA) Method 24."
Hazardous air pollutants (HAP)	"Contains (insert HAP content here in grams per liter and pounds per gallon) solids (non-volatiles) per 40 CFR 63."
Shelf-life	Each unit container, intermediate container where applicable, and shipping container should be marked as follows: "Date of first re-inspection (insert date here that is a minimum of 2 years after date of manufacture)."

6.8 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27 Volume 1, *DoD Shelf-Life Management Program: Program Administration*, and DoDM 4140.27 Volume 2, *DoD Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.9 VOC content certification. When specified (see 6.2), contracting officers will require the manufacturer to provide the Government with a written VOC content certification, as described in 40 CFR 63.785(a)(2), for each batch of corrosion preventive product from which any quantity of the corrosion preventive product is supplied to the Government under the contract. The Government reserves the right to reject as defective any corrosion preventive product for which the VOC content certification has not been provided.

6.10 International standardization agreement implementation. This specification implements STANAG 1135, Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in ASSIST at https://assist.dla.mil.

6.11 Supersession data. This specification supersedes MIL-PRF-16173E(SH) w/INT. AMENDMENT 2, dated 19 October 2017, in its entirety. A cross-reference of current and superseded classification designations is listed in table V.

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TABLE V. Supersession data.

MIL-PRF-16173E classification designations			MIL-PRF-16173F classification designations [1/]	
Grade	**Class**	**Description**	**Class [2/]**	**Grade**
Grade 1	Class I	High VOC, Hard film	Class I	Grade 1
	Class II	Low VOC, Hard film	Class II	Grade 1
Grade 2	Class I	High VOC, Soft film	Class I	Grade 2
				Grade 6
	Class II	Low VOC, Soft film	Class II	Grade 2
				Grade 6
			Class III	Grade 2
				Grade 6
Grade 3	Class I	High VOC, Water displacing, soft film	Class I	Grade 3
				Grade 7
	Class II	Low VOC, Water displacing, soft film	Class II	Grade 3
				Grade 7
			Class III	Grade 3
				Grade 7
Grade 4	Class I	High VOC, Transparent, non-tacky film	Class I	Grade 4
				Grade 7
	Class II	Low VOC, Transparent, non-tacky film	Class II	Grade 4
				Grade 7
Grade 5	Class I	High VOC, Low-pressure steam removable film	Class I	Grade 5
	Class II	Low VOC, Low-pressure steam removable film	Class II	Grade 5
	Class II		Class III	Grade 5

NOTES:

1/ Supersession classifications are based on known product performance. The direction provided in this table is the expected classification but does not override qualification data. Products may qualify for multiple grades but may not qualify for all corresponding grades listed in the supersession table.

2/ Because MIL-PRF-16173F includes new class III requirements for low VOC corrosion preventative compounds, some MIL-PRF-16173E qualified class II products may be qualified to MIL-PRF-16173F as class III products.

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6.12 Subject term (key word) listing.

Adhesion, low-temperature preservative
Miscibility, preservative
Preservative
Preservative, removal
Removable film preservative
Salt spray preventive
Weathering, accelerated

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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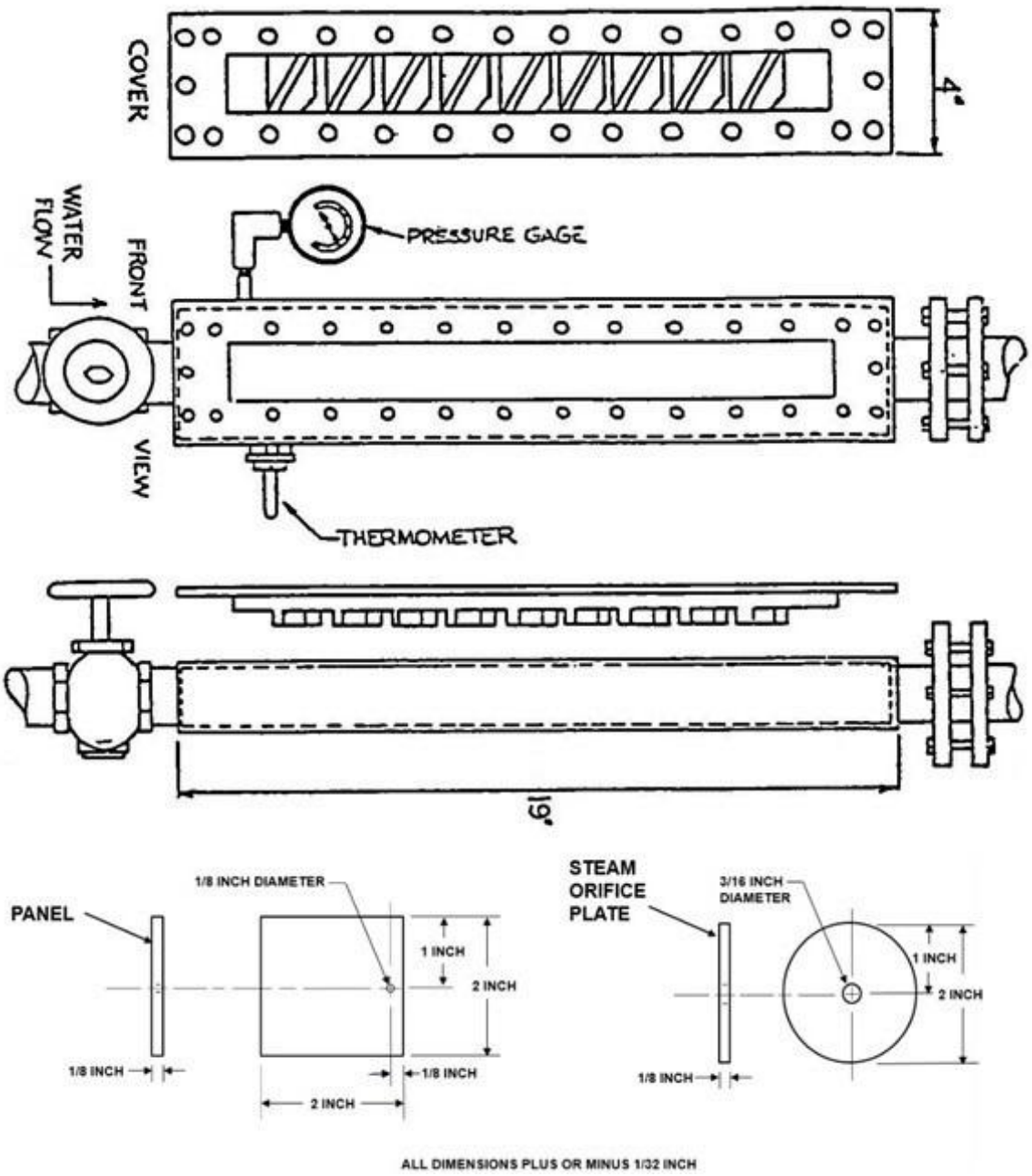


FIGURE 1. Hot water – low-pressure steam removability apparatus.

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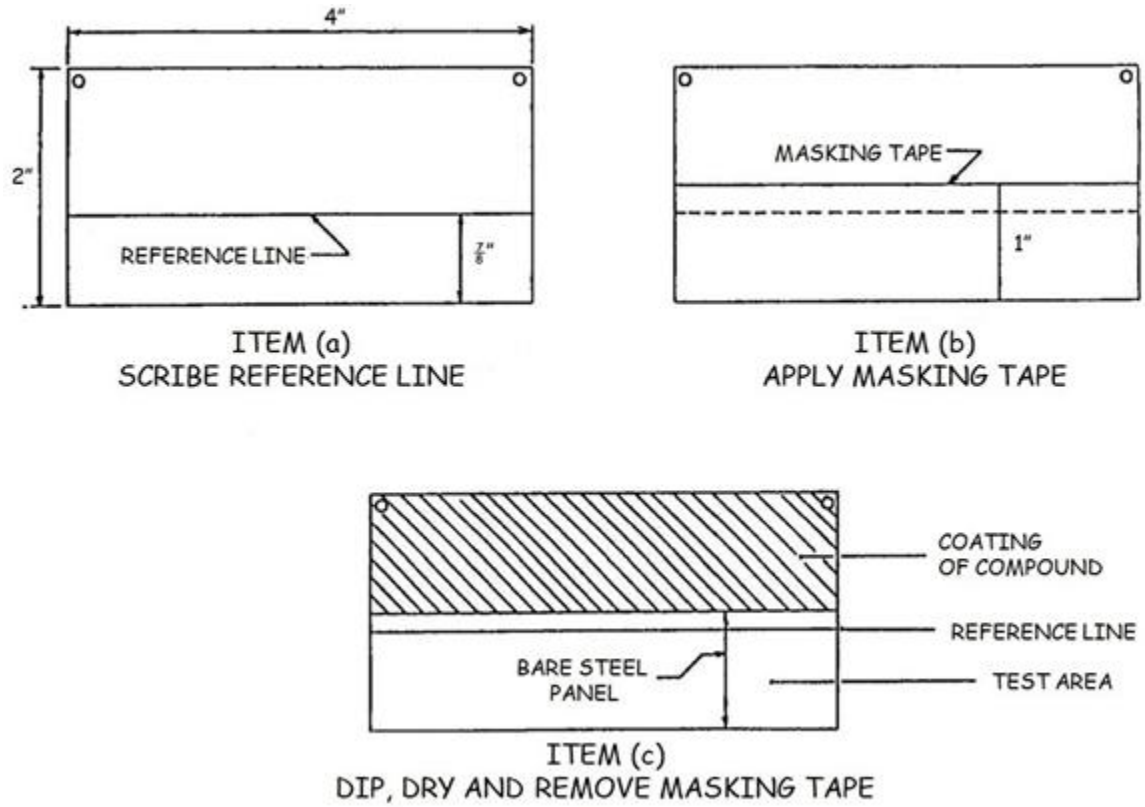


FIGURE 2. Flow resistance test.

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – SH
Air Force – 69

Preparing activity:
Navy – SH
(Project 8030-2025-006)

Review activities:
Army – AR
Navy – AS, MC, OS, YD
Air Force – 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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